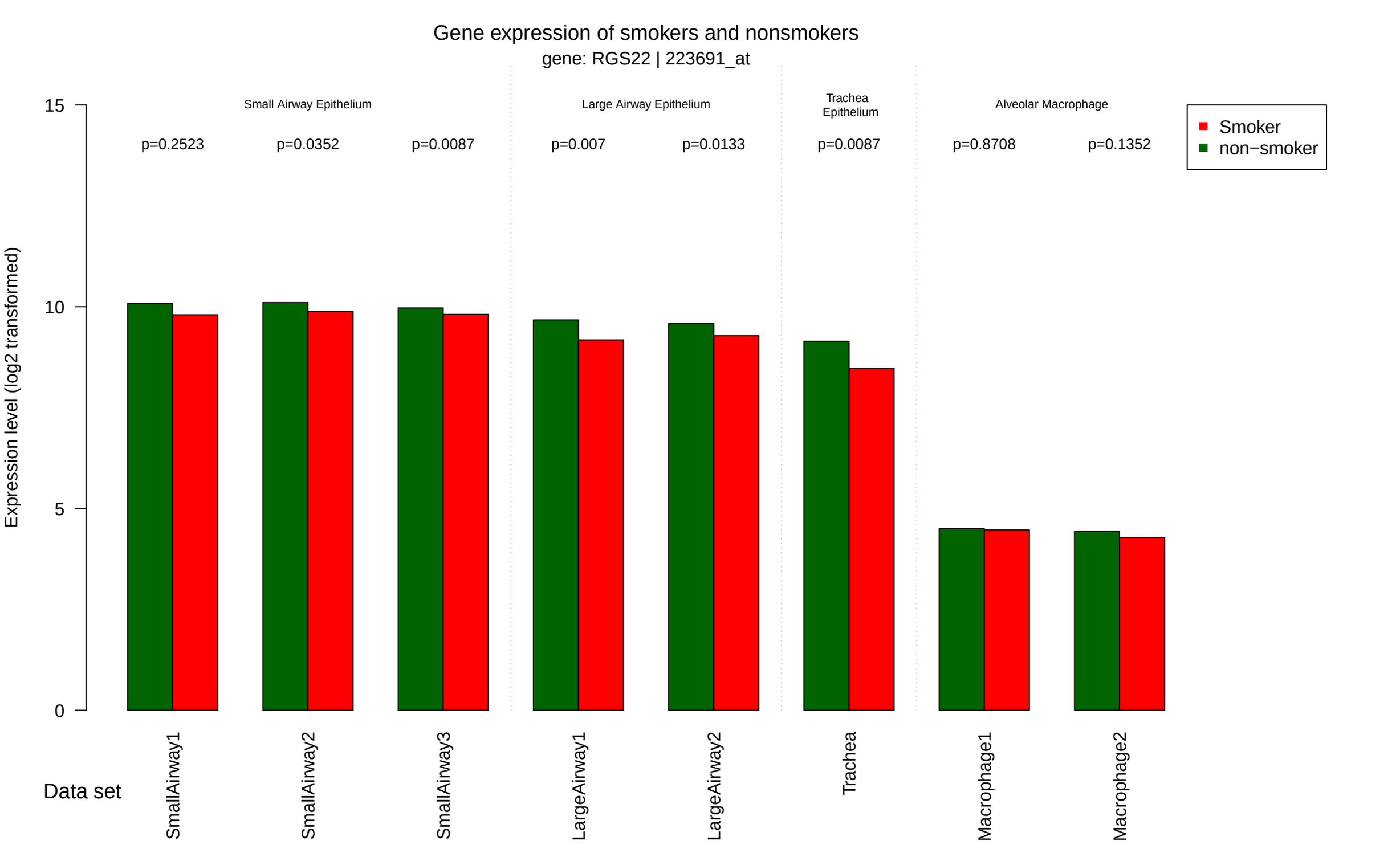
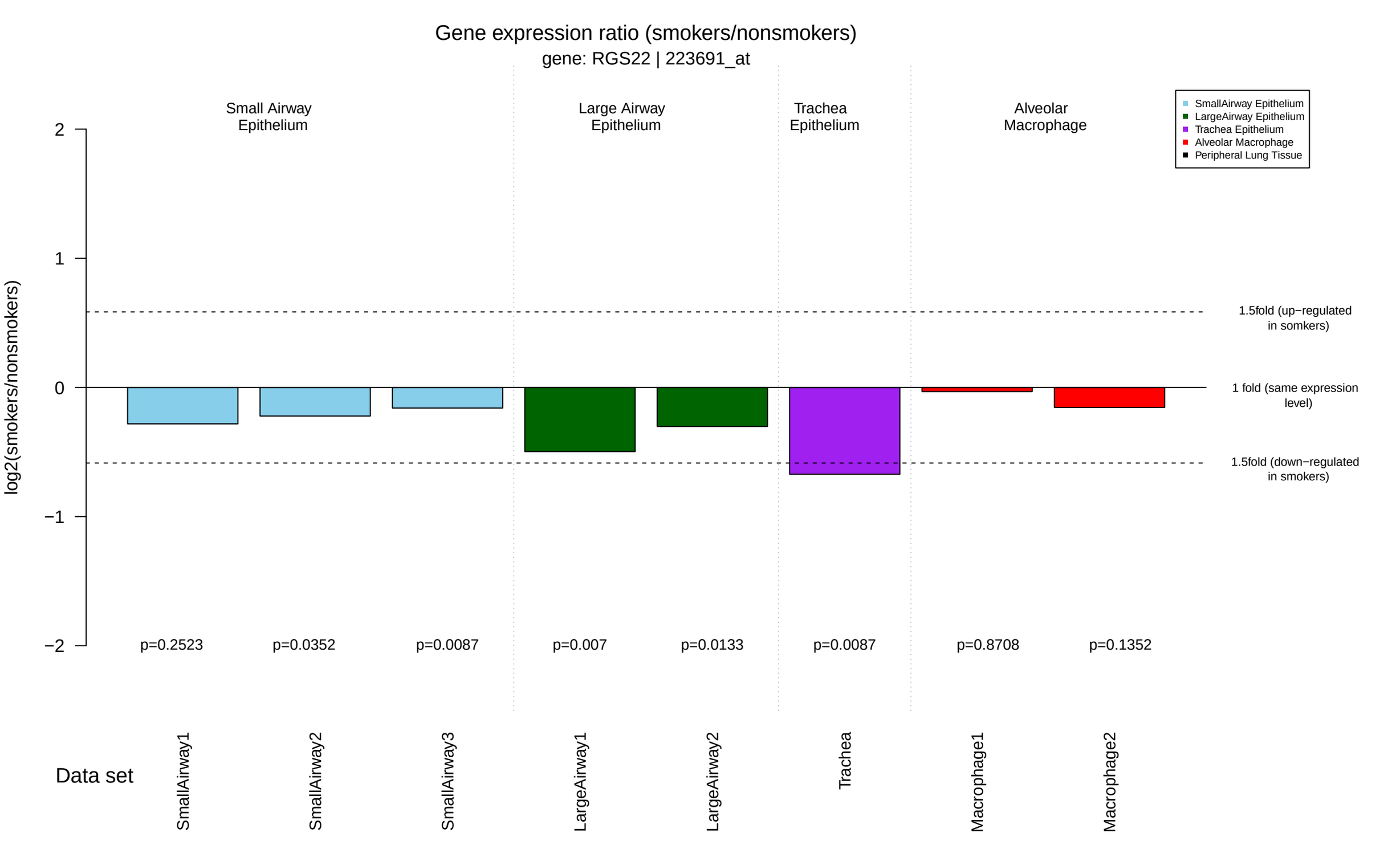
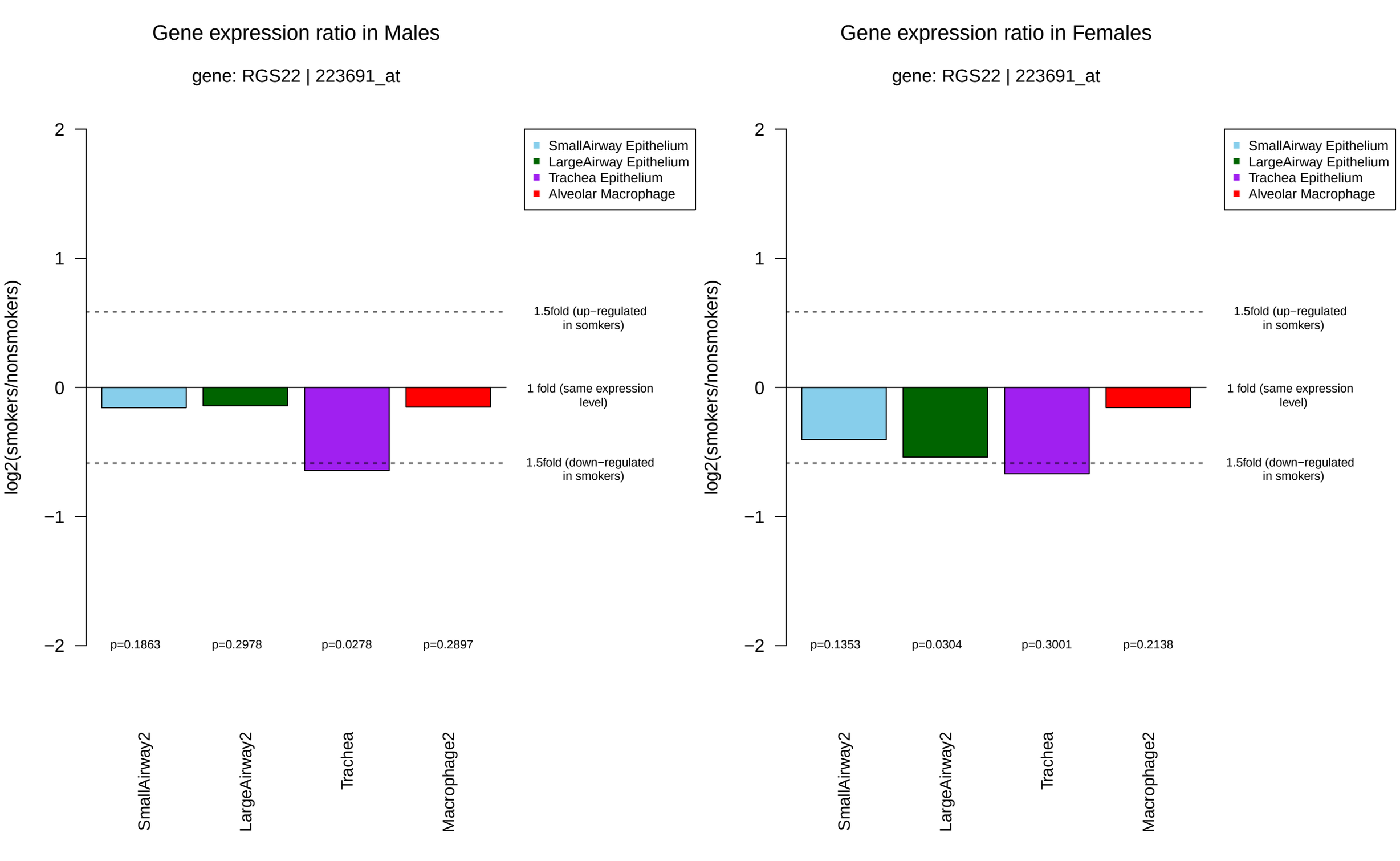
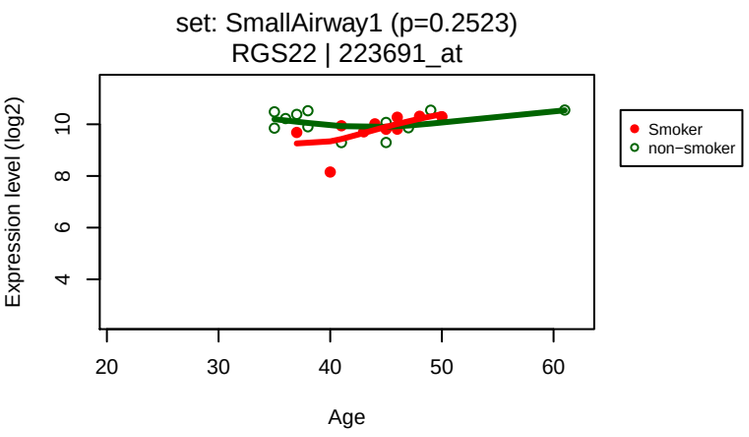
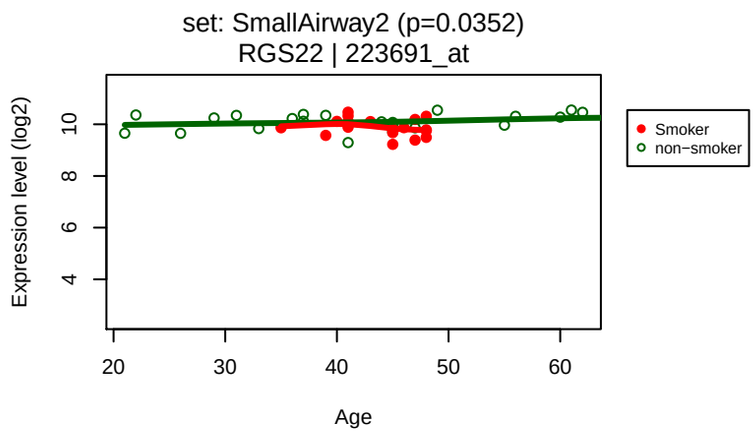
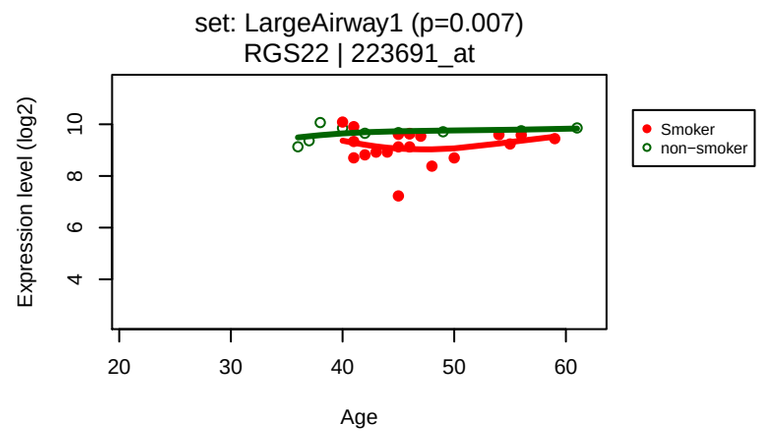

Gene expression of smokers and nonsmokers
gene: RGS22 | 223691_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2523
p=0.0352
p=0.0087
p=0.007
p=0.0133
p=0.0087
p=0.8708
p=0.1352
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

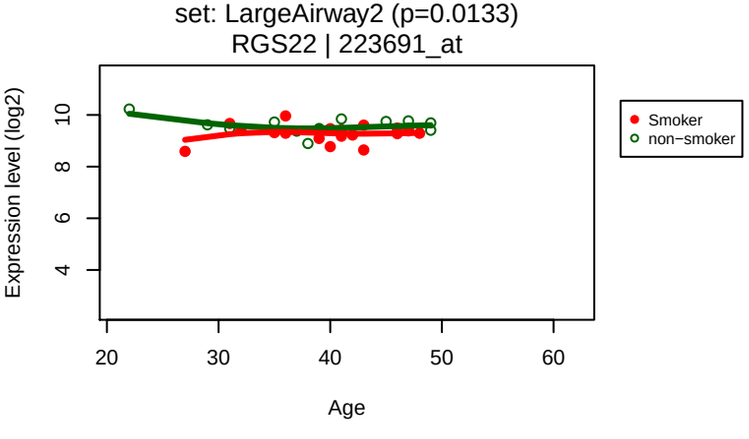

Gene expression ratio (smokers/nonsmokers)
gene: RGS22 | 223691_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2523
p=0.0352
p=0.0087
p=0.007
p=0.0133
p=0.0087
p=0.8708
p=0.1352
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

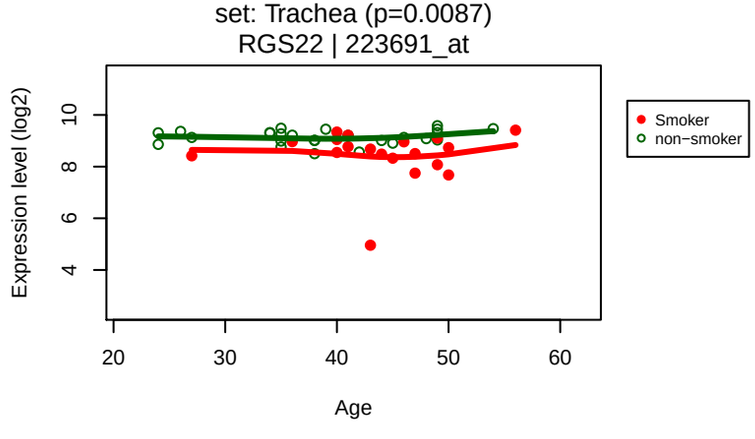

Gene expression ratio in Males
gene: RGS22 | 223691_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1863
p=0.2978
p=0.0278
p=0.2897
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS22 | 223691_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1353
p=0.0304
p=0.3001
p=0.2138
SmallAirway2
LargeAirway2
Trachea
Macrophage2

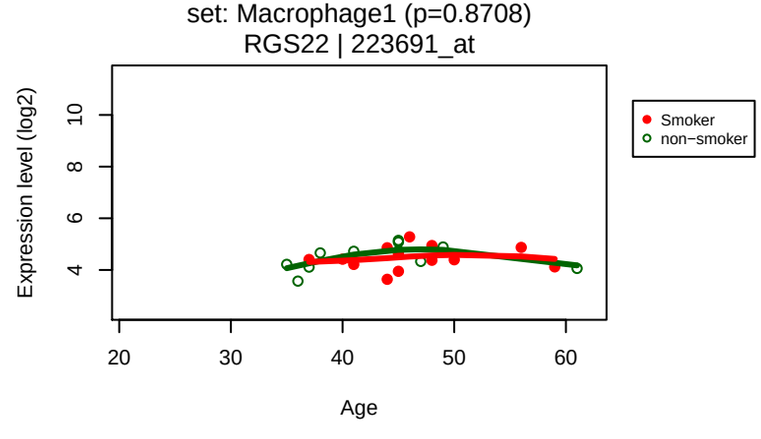

set: SmallAirway1 (p=0.2523)
RGS22 | 223691_at
Expression level (log2)
Age
Smoker
non−smoker

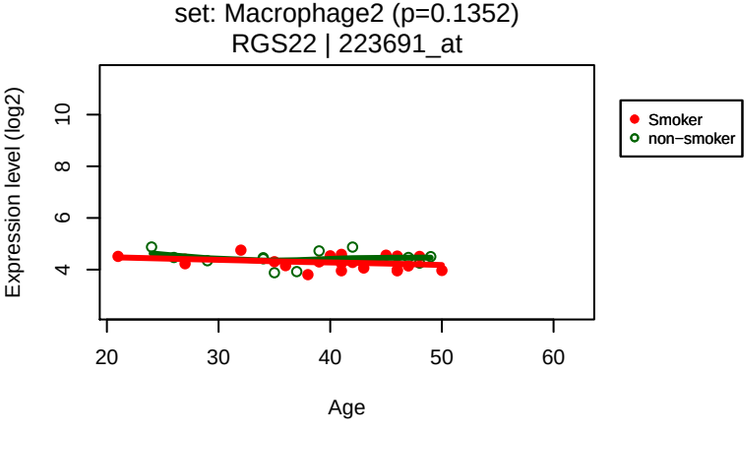

set: SmallAirway2 (p=0.0352)
RGS22 | 223691_at
Expression level (log2)
Age
Smoker
non−smoker

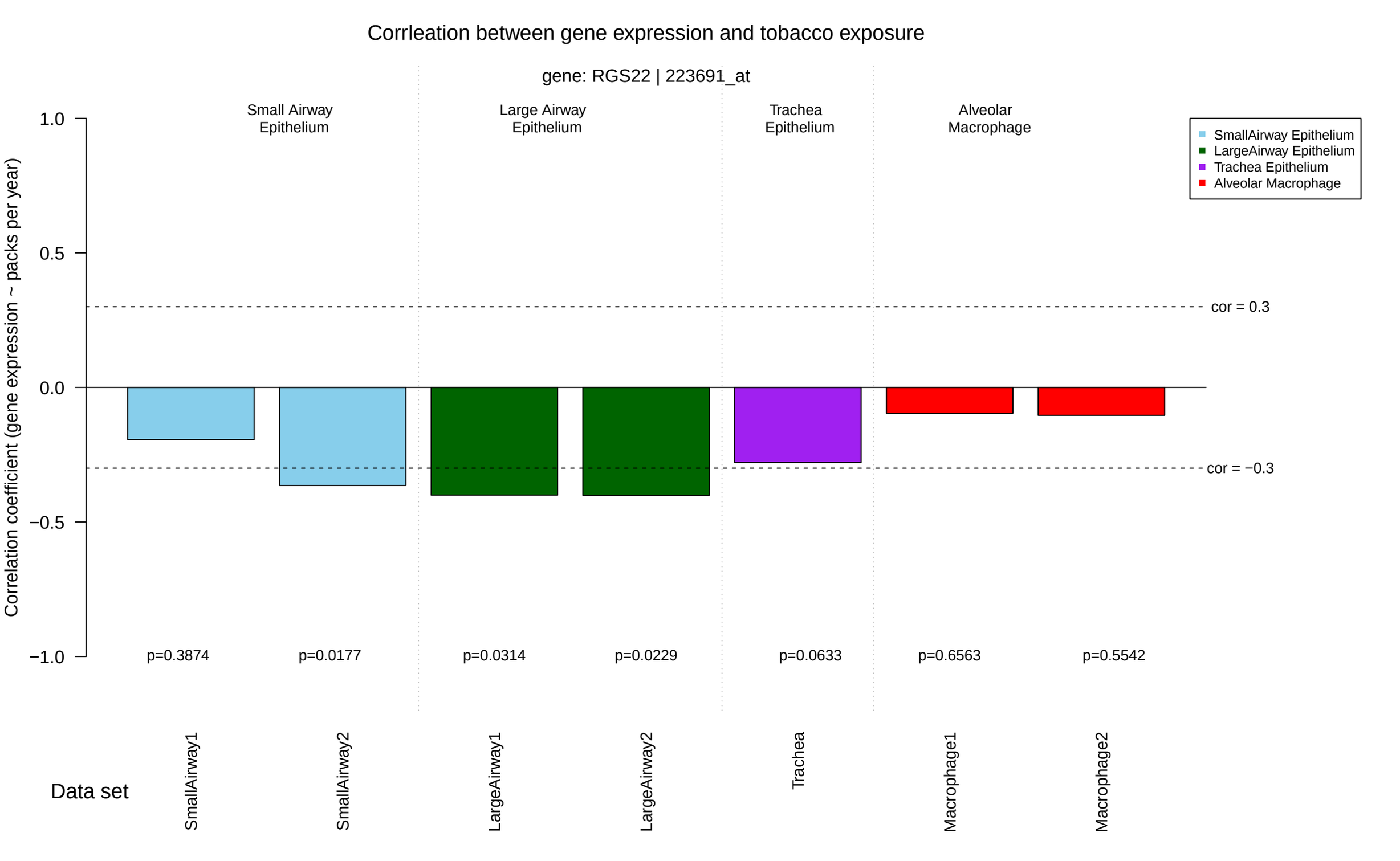

set: LargeAirway1 (p=0.007)
RGS22 | 223691_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0133)
RGS22 | 223691_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0087)
RGS22 | 223691_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8708)
RGS22 | 223691_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1352)
RGS22 | 223691_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS22 | 223691_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3874
p=0.0177
p=0.0314
p=0.0229
p=0.0633
p=0.6563
p=0.5542
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2